Cointegration and Error-Correction Models

These notes accompany Lecture 14. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Lecture 9 warned that regressing one non-stationary series on another can produce a spurious result, and the safe response was to difference everything. But differencing throws away exactly what economists usually care about: the long-run relationship. Cointegration is how we get it back.

Learning outcomes – after this lecture you should be able to:

1. Explain what cointegration is and why a stationary linear combination of I(1) series is special.
2. Describe the error-correction mechanism and identify which variable is adjusting.
3. Write down an ECM and interpret the impact multiplier, the long-run multiplier and the error-correction coefficient.
4. Carry out the four steps of the Engle-Granger procedure, using the correct critical values.
5. State the drawbacks of Engle-Granger and say when FMOLS, DOLS or a VECM is needed.
6. Test weak exogeneity to decide between a single-equation ECM and a system.

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1. The concept of cointegration

1.1 The problem

Plotting US real personal consumption expenditure against real disposable personal income from 1995 shows a clear upward trend in both, and formal unit-root tests confirm each series is I(1). We know from Lecture 9 that using non-stationary variables in a regression risks a spurious result.

But non-stationary series need not be unrelated – something deep may be driving two or more of them together. A scatter of log consumption against log income shows a strong relationship. These are hints, not proof, that a common force is at work.

The obvious fix is to difference both series, which delivers stationarity:

$$\Delta y_t = b_0 + b_1 \Delta x_t + \Delta u_t$$

> **Why differencing is not enough**
>
> This equation gives correct estimates – but only of the SHORT-RUN relationship between the two variables.
>
> If y and x maintain an economically meaningful equilibrium in the long run, differencing discards it entirely.
>
> And economists are usually asking long-run questions. Is there a supply and demand misalignment? Is the exchange rate over- or undervalued? Is monetary policy loose or tight? None of these can be answered from a model in differences alone.

1.2 Linear combinations of I(1) series

The key statistical idea is a knife-edge one. Add two I(1) series together and you generally get another I(1) series: the stochastic trends accumulate rather than cancel. The exception arises when the series share the same stochastic trend – then one particular set of weights cancels it, and what is left is stationary. Nothing guarantees such weights exist; when they do, it is because something is holding the series together.

That something is economics, not statistics. Where a relationship holds in the long run, the gap between its two sides cannot wander freely, and the theory itself is the restriction that makes the combination stationary. The standard examples are purchasing power parity, uncovered interest parity, the permanent income hypothesis, the unbiased forward rate hypothesis, and commodity market arbitrage.

1.3 What cointegration is

Engle and Granger (1987) noted that a linear combination of two or more I(1) series may be stationary, in which case we say the series are **cointegrated**. A set of I(1) series is cointegrated if there exists at least one stationary linear combination of them.

For log consumption c and log income r, cointegration means there exist non-trivial coefficients b_1 and b_2 – not all zero – such that $b_1 c + b_2 r$ is stationary.

If that combination is stationary while each series individually is not, the two cannot drift apart without limit – something has to pull them back. Identifying that something is the rest of the lecture, and the answer is error correction.

1.4 The error-correction mechanism

Write the long-run relationship between log consumption and log income:

$$c_t = \beta_0 + \beta_1 r_t + u_t$$

At any time t you can consume more than your income implies – by taking out a loan – or less, perhaps because you expect financial difficulties in the future. That disequilibrium is captured by u_t: additional debt or additional saving.

But the deviation cannot be infinite. You will have to repay the loan in future, by reducing purchases of other goods and services, and your saving rate will return to its natural level. So consumption will be adjusted back to the level corresponding to income. That is the error-correction mechanism.

1.5 Three ways the adjustment can happen

Drop the error terms and consider the long-run line $c^*_t = \beta_0 + \beta_1 r_t$ with $\beta_1 > 0$. Suppose consumption rises so that $c_t - c^*_t > 0$: a positive gap has opened, and consumption is too high relative to what income suggests. There are three ways the relationship can be restored.

Case	What adjusts	Equation	Sign
1	Consumption does all the work. c^*_t is unchanged and c_t falls	$\Delta c_t = \alpha_c(c_{t-1} - c^*_{t-1})$	$\alpha_c < 0$
2	Income does all the work. c_t is unchanged; r_t and hence c^*_t rise	$\Delta r_t = \alpha_r(c_{t-1} - c^*_{t-1})$	$\alpha_r > 0$
3	Both adjust simultaneously	Both equations together	$\alpha_c < 0$, $\alpha_r > 0$

In every case the change in the adjusting variable this period is proportional to last period's gap: the further the system sits from the long-run line, the harder it is pulled back. Case 3 is the general one, and the system settles at a point between the two extremes rather than at either of them.

> **Reading the error-correction coefficient**
>
> Its SIGN must be corrective – it guarantees that the pull is back toward equilibrium rather than away from it.
>
> Its MAGNITUDE is the speed of return: how much of the gap closes each period.
>
> Whether "corrective" means negative or positive depends on which variable's equation you are looking at, and on how the gap is defined. Always work it out rather than memorising a sign.

Cointegration and error correction are two descriptions of the same thing: if a stationary combination of c and r exists, some adjustment mechanism must be enforcing it. The argument generalises directly to three or more variables.

1.6 The error-correction model

If y_t and x_t are cointegrated, their linear combination $u_t = y_t - \beta_0 - \beta_1 x_t$ is stationary by definition, so $u_t \sim I(0)$. We can therefore write:

$$\Delta y_t = b_0 + b_1 \Delta x_t + \alpha u_{t-1} + e_t, \quad \text{where } \alpha < 0$$

Parameter	Name	Interpretation
b_1	Impact multiplier	The short-run effect – the immediate impact of a change in x on a change in y
β_1	Long-run multiplier	From the level relationship estimated in step 2, not from this equation
α	Error-correction coefficient	The adjustment effect – how much of the previous period's disequilibrium is corrected in period t. Expected to be negative under the convention used here (see the box below)

This has the advantage of including both long-run and short-run information. Lags of Δy_t and Δx_t, or other exogenous variables, can be added.

> **A note on the sign convention – read this carefully**
>
> The equation above is written with a PLUS in front of α, and α is expected to be NEGATIVE. That is the same convention as Section 1.5, where $\Delta c_t = \alpha_c(c_{t-1} - c^*_{t-1})$ with $\alpha_c < 0$, and it is the convention used on the lecture slides and throughout these notes.
>
> Many textbooks instead write $-\alpha u_{t-1}$ and then report α as POSITIVE. The economics is identical; only the bookkeeping differs. Before interpreting any published estimate, check whether the minus sign sits in the equation or has been absorbed into the coefficient.
>
> EViews reports whatever you typed. If you enter the lagged residual directly as a regressor, expect a negative coefficient – and if you get a positive one, the mechanism is pushing the system away from equilibrium rather than back toward it.

1.7 A drunk and her dog

Murray's (1994) illustration is worth remembering. A drunk leaves a bar and her steps are completely random – a random walk, hence I(1). Her dog runs alongside, also chaotically, sniffing the bushes – a second I(1) series.

But the drunk keeps the dog on a leash. They can wander apart, but the leash has a finite length, so the distance between them at any time is I(0). The leash is cointegration. When the dog runs too far ahead the leash tightens and the drunk pulls it back: that is error correction.

1.8 Summary of the concept

- Two unrelated trending variables can have high correlation, but their linear combination is I(1) – that is spurious regression.
- Cointegration is stronger than correlation: not just moving together, but maintaining an economically meaningful equilibrium relation over time.
- Cointegration and error correction are the same fact seen from two sides. Variables can wander individually and still remain tied together, because an adjustment mechanism keeps the gap between them bounded.

2. The error-correction mechanism: simulated data

Four experiments make the mechanism concrete. In each, nrnd denotes independent white noise, and 1,000 observations are simulated starting from zero.

Experiment	Data-generating process	Result
A. Unrelated random walks	x = x(−1) + 0.20 + 1.5·nrnd y = y(−1) + 0.05 + 2.5·nrnd	Both trend upward, but the distance (y − x) has no mean-reverting dynamics. Non-stationary, I(1). No long-run equilibrium
B. Correct error-correction terms	x1 = … + 0.05·(y1(−1) − x1(−1)) y1 = … − 0.05·(y1(−1) − x1(−1))	The two series now move together strongly; the earlier divergence disappears. The distance is stationary with expectation near zero – cointegration
C. Smaller coefficients	Same as B, but ±0.005 instead of ±0.05	Still cointegrated, but co-movement is visibly weaker: the distance is less clearly stationary and its variance is larger. Magnitude governs the speed of convergence
D. Wrong signs	x3 = … − 0.05·(y3(−1) − x3(−1)) y3 = … − 0.05·(y3(−1) − x3(−1))	The distance becomes non-stationary again – no cointegration. The adjustment mechanism does not work

> **The three lessons**
>
> Two independent random walks diverge.
>
> An error-correction model with the CORRECT signs produces a stationary distance and hence cointegration. With the wrong signs it produces a non-stationary distance and no cointegration.
>
> The error-correction coefficient is the speed of return to equilibrium – it shows how much of the gap closes each period. Sign determines whether the system converges at all; magnitude determines how fast.

Note in experiment D that both coefficients carry the same sign. With the gap measured as y − x, the correcting equation needs a negative coefficient and the other a positive one – pulling from opposite directions. Two negatives push the system the same way, so nothing corrects.

3. The Engle-Granger approach

Engle and Granger (1987) introduced a simple test for the existence of a cointegrating relationship, in four steps.

Step 1. Test the order of integration

Cointegration requires the variables to be integrated of the same order, so test each with ADF, PP or KPSS. Three outcomes:

Finding	What to do
Both I(0)	Stop. Estimate a classical regression or a VAR in levels – an I(0) series already reverts to its mean, so there is no stochastic trend to share

Finding	What to do
Different orders	Stop. They cannot be cointegrated
Both the same order, almost always I(1)	Proceed to step 2 – cointegration and an error-correction mechanism may be present

Step 2. Estimate the long-run relationship

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

Estimate by OLS in levels and keep the residuals. If there is no cointegration the results will be spurious. If the variables are cointegrated, OLS yields super-consistent estimators of the cointegrating parameter β_1 – it converges faster than the usual rate.

Step 3. Test whether the residuals are stationary

This is the actual cointegration test. Perform an ADF test on the residual series, and do not include a constant or a time trend – the residuals have mean zero by construction.

> **The critical values are NOT the standard ADF ones**
>
> Because u_t is itself an estimated residual, the critical values for the null of no cointegration are more negative than the standard ADF values.
>
> Use MacKinnon (2010). The values also depend on the number of variables in the long-run relationship – more variables means a more negative critical value.
>
> Using standard ADF critical values here will make you find cointegration far too often.

If $u_t \sim I(0)$ we reject the null that y and x are not cointegrated.

Step 4. Estimate the error-correction model

Regress the change in y on the change in x and the lagged residual, adding lags of Δy or Δx if needed:

$$\Delta y_t = b_0 + b_1 \Delta x_t + \alpha u_{t-1} + e_t, \quad \text{where } \alpha < 0$$

The coefficient on the lagged residual should be statistically significant and carry the correct sign, so that the error-correction mechanism actually works. Then check adequacy with the usual diagnostics: the ECM residuals should be white noise – uncorrelated, homoskedastic, and ideally normally distributed.

Drawbacks of Engle-Granger

The approach is easy to understand and implement, but it has three important shortcomings.

Drawback	Detail
Which variable goes on the left?	The procedure says nothing about it. As the sample goes to infinity, testing the residuals from y-on-x and from x-on-y is equivalent – but economic samples are rarely large, so in practice one regression may exhibit cointegration while the other does not. Use economic theory to decide

Drawback	Detail
Only one cointegrating relationship	With more than two variables there may be several. A procedure based on the residuals from a single relationship cannot handle this. Use the Johansen approach if you suspect multiple relationships.
A two-step estimator	Step one generates the residuals, step two tests them. Any error introduced in the first step is carried into the second.

4. Practical refinements

4.1 FMOLS or DOLS instead of OLS in step 2

In a cointegrating regression OLS gives a consistent β but invalid standard errors. The long-run error term is generally correlated with its own past (serial correlation) and often with the regressors (endogeneity). Neither biases β – it is still super-consistent – but both contaminate its standard error.

Three methods correct this for a single cointegrating vector: Fully Modified OLS (FMOLS, Phillips & Hansen 1990), Dynamic OLS (DOLS, Stock & Watson 1993), and Canonical Cointegrating Regression (CCR, Park 1992). All three fix the standard errors so you can actually test hypotheses about the long-run relationship.

Rule of thumb: use FMOLS or DOLS when you want to test the long-run coefficient, not merely estimate it.

Comparing the three estimates – what to look at

Compare slopes, not intercepts. β is the long-run content; the constant shifts mechanically with β.

The constant is approximately mean(log C) − β·mean(log Y), which makes it extraordinarily sensitive to β. A change in β of 0.03 multiplied by a mean log income around 9.5 moves the implied constant by roughly 0.3.

DOLS also uses fewer observations – it burns them forming leads and lags – so part of any difference is a sample effect rather than a methodological one.

4.2 A VAR instead of a single equation in step 4

A single-equation ECM assumes only one variable adjusts to disequilibrium – it forces you to assume who adjusts. But we cannot be sure. Other variables may also respond to deviations from equilibrium, or all of them may.

A system lets every variable adjust, and lets you test which ones actually do. If several adjust, the single equation is misspecified and only the system captures the full adjustment.

5. Worked example: US consumption and income

Real personal consumption expenditure and real disposable personal income, both in billions of chained 2017 dollars, seasonally adjusted, from FRED.

Steps 1–3

Step	Result
1. Order of integration	ADF confirms both series have a unit root in log-levels and are stationary in log-differences – both I(1), so we may proceed
2. Long-run relationship	OLS in log-levels gives a long-run multiplier close to 1: a 1% increase in disposable income is associated with about a 1% increase in consumption. Expected sign and plausible magnitude – but the regression could still be spurious
3. Cointegration test	ADF on the residuals (no constant, no trend) gives t = −3.89 against a 5% MacKinnon critical value of −3.40. We reject the null of no cointegration at 5%. The residuals show mean-reverting dynamics around zero, with the pandemic producing a large temporary disequilibrium

Step 4 – the ECM

It is reasonable to assume consumption does the adjusting: people cut or increase their spending rather than their incomes changing in response. So we estimate the ECM for first differences with the lagged residual added, initially on a sample shortened to avoid the COVID distortion.

Specification	Error-correction coefficient	Note
Baseline ECM	Correct sign, significant – about 10% of the gap closes per period	If consumption grows above the equilibrium level it falls the next period. But explanatory power is low
With lags and a Lehman dummy	Significant at 5%, about 6% of the gap closes per period	Lags capture persistence from habit formation and consumption smoothing. The slow adjustment reflects households smoothing consumption rather than jumping back to the line
Full sample with COVID dummies	Broadly unchanged	Impulse dummies for the 2020 lockdowns and transfers, plus a level-shift dummy for the post-2019 change in the smoothing pattern

Diagnostics on the final specification: we fail to reject no residual autocorrelation, fail to reject homoskedasticity, and fail to reject normality. The model is adequate and can be used for testing hypotheses or forecasting.

Testing weak exogeneity in a system

If you are unsure which variable adjusts, estimate the system rather than a single equation. Specify the VAR in first differences and add the lagged residual from the long-run regression to every equation as an exogenous regressor. Each equation then reports its own error-correction coefficient, and you can test which variables actually respond.

A word on terminology. This is a two-step system ECM: the cointegrating vector is imposed from step 2 rather than estimated alongside the adjustment coefficients. A VECM in the strict sense estimates both jointly by maximum likelihood, which is what the Johansen procedure adds.

Equation	Error-correction coefficient	Conclusion
Consumption	Correct sign, significant at 5%	Consumption adjusts
Income	Correct sign but not significant	Income does not react to deviations from the long-run relationship – it is weakly exogenous

So consumption does all the work, and the system confirms that the single-equation ECM was the right choice. This is Case 1 from Section 1.5 – established by testing rather than assumed.

Refining further – adding lags of the endogenous variables and the Lehman Brothers collapse dummy on the basis of information criteria and diagnostics – leaves the residuals uncorrelated and homoskedastic, with the VAR stationary.

Testing cointegration from the Equation object

If you estimated the long-run relationship by DOLS or FMOLS you can test for cointegration directly from the equation object. Note what the test actually uses. A residual-based cointegration test is a unit-root test in disguise: Engle-Granger and Phillips-Ouliaris both take the residuals from the static OLS long-run regression – not the FMOLS or DOLS residuals, even when launched from an FMOLS equation – and ask whether they contain a unit root. Rejecting non-stationarity in those residuals is what "cointegration" means operationally.

6. Summary

6.1 The decision tree

Begin with the raw series and their graphs, using both intuition and unit-root tests to judge whether each is I(1) or I(0) – and watching for I(2).

Situation	What to estimate
All variables I(0)	Single regression, or a VAR in levels with an appropriate lag order
Two or more I(1)	Run the cointegration test: long-run regression, then ADF on the residuals. Prefer FMOLS or DOLS if you want to test hypotheses about the long-run multipliers
No cointegration	Single regression, or a VAR in differences
Cointegration found	A system with the disequilibrium term in every equation. Then test weak exogeneity
Only one variable adjusts	A single-equation ECM in differences is sufficient
Several variables adjust	Stay with the VECM

A mix of I(1) and I(2) variables requires methods beyond this course. The cointegration test itself is unavoidable – it is what stands between a long-run relationship and a spurious one.

6.2 Advantages of the ECM

1. It measures the correction from the previous period's disequilibrium, which has a clear economic interpretation.

2. Given cointegration, it is formulated in first differences, which typically eliminates trends and resolves the spurious regression problem – while still retaining the long-run information through the error-correction term.
3. It sits naturally inside a general-to-specific modelling strategy: start with a generously parameterised ECM and pare it back to the smallest specification the diagnostics still accept.
4. The disequilibrium error term is stationary by the definition of cointegration. This implies there is an adjustment process preventing errors in the long-run from growing without bound.

> **What this lecture resolved**
>
> Lecture 10 left us with a dilemma: regress in levels and risk a spurious result, or difference and lose the long run.
>
> Cointegration dissolves it. If a stationary linear combination exists, the levels relationship is real and can be estimated – and the ECM puts the short run and the long run in the same equation.
>
> The price of admission is the test in step 3. Everything downstream depends on it.

Key terms

Term	Meaning
Cointegration	A set of I(1) series is cointegrated if at least one linear combination of them is stationary.
Cointegrating vector	The coefficients of that stationary linear combination.
Error-correction mechanism	The adjustment by which variables respond to deviations from the long-run relationship.
Error-correction coefficient	The coefficient on the lagged disequilibrium; its magnitude is the speed of adjustment and its sign must be corrective – negative under the convention used in these notes.
Impact multiplier	b_1 – the immediate short-run effect of a change in x on a change in y.
Long-run multiplier	β_1 – from the levels relationship, not from the ECM.
Super-consistency	OLS on a cointegrating regression converges to β faster than the usual rate.
Engle-Granger procedure	Four steps: order of integration, long-run regression, ADF on residuals, then the ECM.
MacKinnon critical values	The correct critical values for a residual-based cointegration test; more negative than standard ADF values, and dependent on the number of variables.
FMOLS / DOLS / CCR	Methods that correct the standard errors of a cointegrating regression so the long-run coefficient can be tested.
VECM	Vector error-correction model – a VAR in differences with the disequilibrium term in every equation. In the strict sense the cointegrating vector is estimated jointly with the adjustment coefficients (Johansen); the two-step version of Section 5.1 imposes it from the long-run regression instead.
Weak exogeneity	A variable is weakly exogenous if it does not respond to deviations from the long-run relationship.
Johansen procedure	A system method able to detect and estimate more than one cointegrating relationship.

Review questions

Use these to check your understanding before the seminar.

1. Why is differencing both series not a satisfactory response to non-stationarity? Give two economic questions it cannot answer.
2. Explain why a stationary linear combination of I(1) series is "special". Give one economic theory that would predict one.
3. Define cointegration precisely. Why must the coefficients be non-trivial?
4. In the consumption example, what does the error term u_t represent economically, and why can it not grow without bound?
5. Set out the three adjustment cases. For each, state which variable moves and the sign its coefficient must take.
6. Write the ECM. Identify the impact multiplier, the long-run multiplier and the error-correction coefficient, and say where each comes from.
7. In simulation D both coefficients are negative and cointegration disappears. Explain why opposite signs are needed when the gap is defined as $y - x$.
8. Halving the error-correction coefficients does not destroy cointegration but changes something. What, and why?
9. Your two variables are I(1) and I(2). What do you conclude, and at which step of Engle-Granger?
10. Why must the ADF test in step 3 exclude a constant and a trend?
11. State the three drawbacks of Engle-Granger. Which is most serious with four variables, and what would you use instead?
12. When would you choose DOLS over OLS in step 2? What do you give up?
13. Why should you compare slopes rather than intercepts across OLS, FMOLS and DOLS estimates?
14. In the worked example, $t = -3.89$ against a critical value of -3.40. State the conclusion and what would follow if the test had failed.
15. The error-correction coefficient falls from about 10% to about 6% when lags are added. Interpret both figures and explain why the second is economically plausible.
16. The income equation in the system has the correct sign but is insignificant. What does this establish, and what does it license you to do?
17. Trace the decision tree from raw data to a final model for the case of two I(1) variables that turn out to be cointegrated with only one adjusting.

References and further reading

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 17.

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapter 18-4.

Engle, R. F., & Granger, C. W. J. (1987). Co-Integration and Error Correction: Representation, Estimation, and Testing. Econometrica, 55(2), 251–276.

Murray, M. P. (1994). A Drunk and Her Dog: An Illustration of Cointegration and Error Correction. The American Statistician, 48(1), 37–39.

MacKinnon, J. G. (2010). Critical Values for Cointegration Tests. Queen's Economics Department Working Paper No. 1227.

Enders, W. (2015). Applied Econometric Time Series, 4th ed. Wiley – chapter 6.

Macroeconometric Forecasting. IMF online course. URL: https://www.edx.org/course/macroeconometric-forecasting-2.

Park, J. Y. (1992). Canonical Cointegrating Regressions. Econometrica, 60(1), 119–143.

Phillips, P. C. B., & Hansen, B. E. (1990). Statistical Inference in Instrumental Variables Regression with I(1) Processes. Review of Economic Studies, 57(1), 99–125.

Stock, J. H., & Watson, M. W. (1993). A Simple Estimator of Cointegrating Vectors in Higher Order Integrated Systems. Econometrica, 61(4), 783–820.